

## HERMOSILLO, Mexico

WMO: 761600

Lat: 29.0784N

Lon: 110.9310W

Elev: 692

StdP: 14.33

Time Zone: -7.00 (W07)

Period: 00-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 40.9       | 44.2 | 10.6                          | 9.7 | 73.1 | 17.2 | 13.5 | 72.3 | 18.0                  | 68.9 | 15.2 | 69.9 | 1.0                   | 30   | 0.389 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.8                   | 109.1           | 72.9 | 107.0 | 73.5 | 104.4 | 73.1 | 80.8                | 94.6 | 79.9 | 93.8 | 79.1 | 93.1 | 7.9                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.4                            | 146.4 | 85.4 | 76.9 | 143.9 | 85.3 | 75.5 | 137.0 | 85.5 | 44.9          | 95.7 | 44.0 | 94.0 | 43.1 | 93.0 | 87.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 18.0          | 15.3 | 13.0 | 35.4                       | 115.2 | 3.4                | 2.9 | 33.0                                               | 117.2 | 31.1       | 118.9 | 29.2       | 120.6 | 26.7       | 122.7 |    |    |
| (5)               |      |      | 30.0                       | 84.9  | 2.9                | 1.6 | 27.9                                               | 86.1  | 26.2       | 87.0  | 24.6       | 87.9  | 22.5       | 89.0  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 77.6   | 62.4  | 65.2  | 70.6  | 76.0  | 82.0  | 90.5  | 92.3  | 91.1  | 88.2  | 80.0  | 70.7  | 62.0  |  |  |
|                                                                        | DBStd       | 12.03  | 5.51  | 5.94  | 5.76  | 5.05  | 4.94  | 4.09  | 4.03  | 3.93  | 4.02  | 5.64  | 5.99  | 5.18  |  |  |
|                                                                        | HDD50       | 2      | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD65       | 339    | 114   | 66    | 18    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 20    | 119   |  |  |
|                                                                        | CDD50       | 10090  | 386   | 425   | 638   | 779   | 993   | 1214  | 1310  | 1275  | 1146  | 930   | 620   | 374   |  |  |
|                                                                        | CDD65       | 4951   | 33    | 70    | 191   | 330   | 528   | 764   | 845   | 810   | 696   | 466   | 190   | 28    |  |  |
|                                                                        | CDH74       | 71335  | 737   | 1160  | 2808  | 4464  | 7320  | 11716 | 12910 | 12032 | 9643  | 5710  | 2286  | 549   |  |  |
|                                                                        | CDH80       | 43439  | 202   | 459   | 1375  | 2511  | 4587  | 7917  | 8531  | 7698  | 5762  | 3204  | 1053  | 140   |  |  |
| (14) Wind                                                              | WSAvg       | 3.9    | 2.7   | 3.3   | 4.0   | 4.7   | 5.1   | 5.5   | 5.5   | 4.3   | 3.6   | 3.0   | 2.8   | 2.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 11.2   | 0.9   | 0.5   | 0.2   | 0.1   | 0.1   | 0.2   | 3.1   | 3.3   | 1.5   | 0.5   | 0.3   | 0.7   |  |  |
|                                                                        | PrecMax     | 19.1   | 3.9   | 3.6   | 1.1   | 1.1   | 0.9   | 1.9   | 6.1   | 6.5   | 4.4   | 2.0   | 1.9   | 3.9   |  |  |
|                                                                        | PrecMin     | 7.3    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.8   | 0.8   | 0.1   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 2.8    | 1.1   | 0.9   | 0.3   | 0.3   | 0.2   | 0.5   | 1.3   | 1.3   | 1.1   | 0.6   | 0.4   | 1.0   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 88.2  | 95.0  | 99.0  | 102.4 | 107.3 | 112.7 | 111.5 | 109.6 | 107.9 | 104.3 | 97.1  | 88.1  |  |  |
|                                                                        |             | MCWB   | 59.3  | 60.4  | 62.3  | 63.7  | 66.5  | 70.3  | 75.6  | 75.9  | 74.4  | 69.0  | 64.2  | 59.7  |  |  |
|                                                                        | 2%          | DB     | 85.1  | 89.9  | 95.0  | 98.9  | 103.9 | 109.2 | 108.0 | 107.2 | 105.4 | 100.7 | 93.3  | 84.1  |  |  |
|                                                                        |             | MCWB   | 58.1  | 59.7  | 61.0  | 62.8  | 66.4  | 70.5  | 75.0  | 76.5  | 75.1  | 69.6  | 63.3  | 59.1  |  |  |
|                                                                        | 5%          | DB     | 82.2  | 86.0  | 91.5  | 96.4  | 100.7 | 106.2 | 105.9 | 104.2 | 102.3 | 98.3  | 89.9  | 80.4  |  |  |
|                                                                        |             | MCWB   | 57.4  | 58.7  | 59.8  | 62.1  | 65.5  | 71.1  | 75.4  | 76.8  | 74.9  | 69.1  | 63.0  | 57.8  |  |  |
|                                                                        | 10%         | DB     | 78.5  | 82.0  | 88.0  | 93.0  | 98.5  | 104.2 | 103.6 | 102.0 | 99.9  | 94.8  | 86.3  | 76.7  |  |  |
|                                                                        |             | MCWB   | 56.5  | 57.5  | 58.9  | 61.3  | 64.6  | 70.9  | 75.2  | 76.9  | 75.4  | 68.2  | 61.5  | 56.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 64.4  | 65.1  | 65.3  | 67.7  | 71.9  | 79.4  | 82.0  | 82.6  | 81.1  | 78.2  | 70.4  | 64.4  |  |  |
|                                                                        |             | MCDB   | 77.4  | 78.6  | 88.9  | 94.0  | 98.0  | 100.5 | 100.1 | 94.8  | 93.8  | 89.4  | 81.2  | 75.0  |  |  |
|                                                                        | 2%          | WB     | 61.6  | 62.9  | 63.2  | 65.6  | 69.7  | 77.3  | 80.0  | 81.0  | 80.1  | 75.9  | 67.7  | 62.1  |  |  |
|                                                                        |             | MCDB   | 75.6  | 79.5  | 87.0  | 91.2  | 94.9  | 98.2  | 95.7  | 94.2  | 93.0  | 89.9  | 83.8  | 75.3  |  |  |
|                                                                        | 5%          | WB     | 59.8  | 61.2  | 61.7  | 64.1  | 67.9  | 76.0  | 79.0  | 80.1  | 79.2  | 74.1  | 65.4  | 60.5  |  |  |
|                                                                        |             | MCDB   | 75.3  | 78.3  | 86.6  | 89.6  | 93.9  | 96.1  | 94.0  | 93.8  | 92.0  | 88.3  | 81.9  | 74.3  |  |  |
|                                                                        | 10%         | WB     | 58.0  | 59.4  | 60.2  | 62.5  | 66.2  | 74.7  | 78.2  | 79.4  | 78.5  | 72.1  | 63.6  | 58.7  |  |  |
|                                                                        |             | MCDB   | 74.3  | 77.1  | 84.8  | 88.7  | 93.7  | 94.7  | 92.9  | 93.3  | 91.1  | 86.7  | 80.7  | 73.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 30.6  | 30.0  | 32.1  | 32.0  | 31.0  | 28.4  | 21.8  | 21.2  | 22.4  | 27.5  | 29.2  | 28.6  |  |  |
|                                                                        |             | MCDBR  | 35.5  | 37.6  | 37.0  | 36.6  | 33.8  | 30.2  | 24.8  | 24.1  | 26.7  | 31.0  | 33.3  | 34.2  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.8  | 15.6  | 14.7  | 14.4  | 13.3  | 11.0  | 8.3   | 7.8   | 8.6   | 11.0  | 13.5  | 15.5  |  |  |
|                                                                        |             | MCWBR  | 28.7  | 29.1  | 32.9  | 31.7  | 30.3  | 25.4  | 21.0  | 21.3  | 21.1  | 22.6  | 27.8  | 27.5  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.291 | 0.294 | 0.321 | 0.326 | 0.344 | 0.390 | 0.434 | 0.422 | 0.394 | 0.341 | 0.301 | 0.291 |  |  |
|                                                                        | taud        |        | 2.537 | 2.530 | 2.432 | 2.416 | 2.385 | 2.315 | 2.311 | 2.348 | 2.387 | 2.474 | 2.532 | 2.545 |  |  |
|                                                                        | Ebn at Noon |        | 304   | 312   | 307   | 307   | 299   | 284   | 271   | 274   | 279   | 290   | 297   | 298   |  |  |
|                                                                        | Edh at Noon |        | 28    | 30    | 36    | 37    | 39    | 41    | 42    | 40    | 37    | 32    | 27    | 26    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1279  | 1564  | 2021  | 2376  | 2614  | 2553  | 2233  | 2116  | 1898  | 1682  | 1368  | 1176  |  |  |
|                                                                        | RadStd      |        | 74    | 98    | 84    | 79    | 54    | 111   | 90    | 61    | 83    | 81    | 61    | 47    |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page